

ZoikoMeds Conversational Agent — Wireframe & Implementation Specification

Document class: Tier-0 build specification · Engineering-ready

Owner roles: Strategy · Product · Design · CTO · AI

Status: v1.0 — for engineering implementation

Canonical dependencies: MediBase™ ·

ZoikoAvail™ · ZoikoSignal™ · Trust Center ([△](#) unbuilt — see §9)

0. Critical Analysis Before Specification

Before the wireframe, the flaws this specification is engineered to avoid. These are

the failure modes of virtually every healthcare-site chatbot, including those shipped by companies far larger than ZoikoMeds:

1. Strategic flaw — the generic FAQ widget.

Most healthcare chatbots are support-deflection tools bolted onto the corner of a page. That positioning is beneath the platform. On a medicine availability intelligence platform, the agent must be a *front-end to the intelligence layer itself* — a conversational surface over ZoikoAvail™ — or it should not exist.

2. Structural flaw — one bot, four audiences.

The site serves patients, pharmacies, enterprises, and wholesale partners with radically different intents, vocabularies, and compliance postures. A single undifferentiated conversation flow fails all four. The agent must route by persona within the first exchange.

3. Trust flaw — unsourced answers in a

regulated domain. A hallucinated availability claim ("yes, that medicine is in stock") is not a UX bug; it is a patient-safety

event and a legal liability. Every availability response must be grounded, confidence-scored, and timestamped — inheriting the ZoikoAvail™ Bayesian confidence model rather than inventing a parallel truth.

4. **Compliance flaw — the medical-advice**

drift. Patients will ask dosage, interaction, and symptom questions. The agent must have a hard, non-negotiable refusal boundary with a warm handoff to pharmacists/clinicians — enforced at the model layer, not just the prompt layer.

5. **Voice flaw — mascot energy.**

Cutesy avatars, exclamation marks, and "Hi! I'm here to help! 😊" destroy institutional credibility. The register is Stripe support meets Mayo Clinic Digital: precise, calm, humane.

6. **Governance flaw — orphaned disclosure.**

AI disclosure, data-handling, and escalation policies must resolve to one canonical home. Today that home (the Trust Center) does not exist. This is a **blocking dependency**, consistent with the cross-

document governance gap already identified.

Everything below is designed against these six failure modes.

1. Agent Naming Decision

1.1 Recommended name: Zoi™

Formal designation: *Zoi™ – the ZoikoMeds Availability Assistant*

In-product label: Zoi (launcher, header, message attribution)

First-use disclosure line: "Zoi is ZoikoMeds' AI assistant. Zoi does not provide medical advice."

Rationale (contextual, not decorative):

- **Etymology is the strategy.** *Zoi* derives from the Greek ζωή (zoē) – "life." For a platform whose mission is keeping medicines available to the people who need them, the name *is* the mission statement. No other

candidate carries semantic payload this precise.

- **Brand-system coherence.** It is visibly derived from *Zoiko*, sitting naturally beside ZoikoAvail™, ZoikoSignal™, and MediBase™ — but deliberately drops the compound-trademark pattern. Products get compound names; the human-facing agent gets a human-scale name. That distinction is itself Tier-0 design (compare: Stripe's products are "Radar" and "Atlas," but its docs assistant is simply named, small, and quiet).
- **Global pronunciation.** Two phonemes, pronounceable in every major market, no negative homophones in major languages, three characters — ideal for a launcher label at 14px.
- **Trademark posture.** Distinctive, non-descriptive, and defensible within the existing Zoiko Group family filing strategy.

1.2 Candidates evaluated and rejected

Candidate	Rejection rationale
ZoikoAssist™	Generic compound; reads as a feature, not an agent; downmarket.
Avail	Direct collision with ZoikoAvail™; dilutes the product trademark.
MediBot / ZoikoBot	Category-killer generic; "bot" suffix signals low capability and undermines the intelligence positioning.
Ziko	Diminutive of the parent brand; risks infantilising Zoiko Group itself.
Signa / Sig	"Sig" has a specific pharmacy meaning (prescription directions) – dangerous semantic overload in this domain.

2. Strategic Positioning

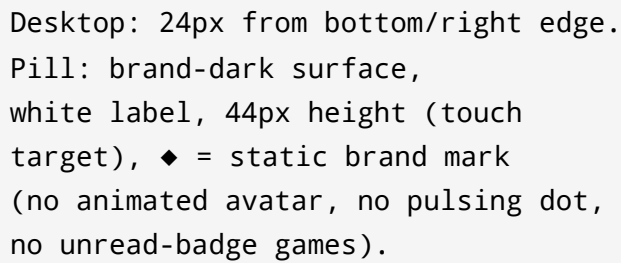
What Zoi is: a conversational surface over the ZoikoMeds intelligence stack — availability lookups (ZoikoAvail™), navigation and explanation of the platform, persona-appropriate onboarding, and structured escalation to humans.

What Zoi is not: a diagnostic tool, a source of medical advice, a prescription service, or a substitute for a pharmacist. This boundary is stated in the UI, enforced in the model, and auditable in logs.

North-star interaction: a caregiver types the name of a medicine and a city, and within one exchange receives a grounded, confidence-scored, timestamped availability answer with a path to a stocking pharmacy — the platform's core value delivered without a single page navigation.

3. Wireframes

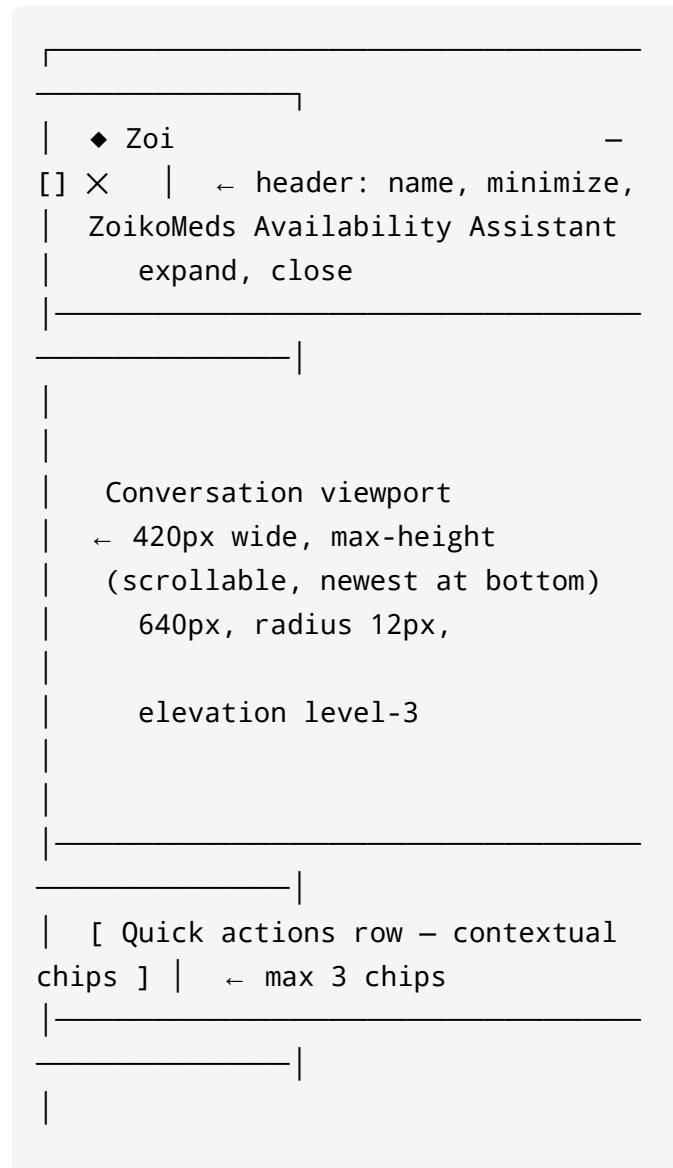
3.1 Launcher (persistent, all pages)

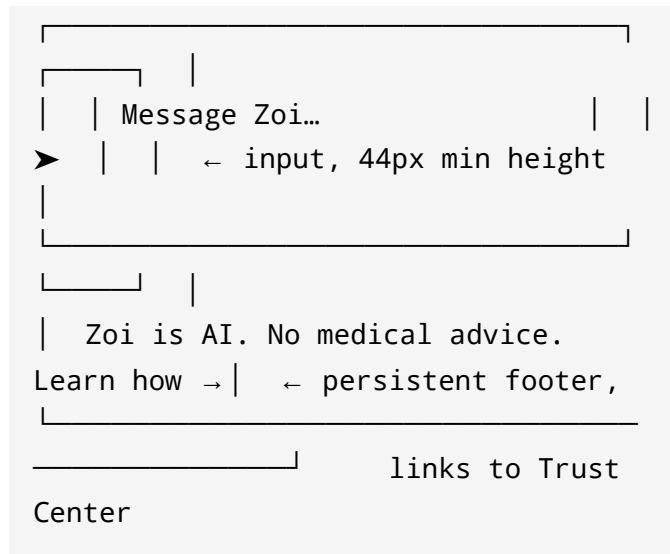


default → "◆ Ask Zoi"
page-aware → contextual variant
(see 3.6), e.g. on a
medicine detail
page: "◆ Check availability"
minimized → ◆ icon only (user-
collapsed, persists in session)

Design rule: the launcher never auto-opens, never plays sound, never shows fake "typing" to bait engagement. Institutional surfaces do not chase clicks.

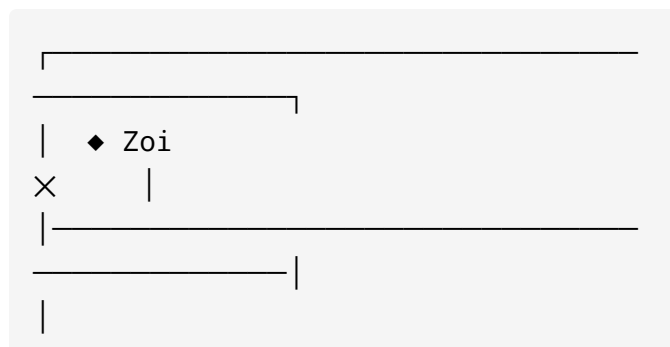
3.2 Panel anatomy – desktop (open state)





Mobile: full-height bottom sheet (100dvh minus safe areas), drag-handle to dismiss, input pinned above keyboard. No floating panel on <768px.

3.3 First-open state (disclosure + persona routing)



|
| Hello. I'm Zoi, the ZoikoMeds
|
| assistant. I can check medicine
|
| availability, explain the
platform, |
| and connect you with our team.
|
|
|

| I'm an AI assistant and don't
give |
| medical advice. For clinical
|
| questions, always consult your
|
| pharmacist or doctor.
|
|
|

| What brings you here today?
|
|
|
|

| | ○ I'm looking for a medicine

```
| | ← Patient/Caregiver  
| | ○ I run or work at a pharmacy  
| | ← Pharmacy  
| | ○ I represent an organisation  
| | ← Enterprise  
| | ○ I'm a wholesale partner  
| | ← Wholesale  
| | ○ Something else  
|  
  
|  
└───────────────────┘  
  
┌──┐ |  
│ ───────────────────┘  
└──────────┬───┘  
|  
  
┌──────────────────────────────────┐  
├──┤ |  
| Message Zoi... | |  
► | |  
|  
└──────────────────────────────────┘  
  
┌──┐ |  
└──────────────────────────────────┘
```

Rules: free-text always works – the persona chips are an accelerator, never a gate. Persona is also inferred from entry page (a user

opening Zoi from the Wholesale Portal skips this step; see 3.6). Disclosure copy appears once per session, never repeated mid-conversation.

3.4 The signature component: the Availability Card

This is the one element Zoi will be remembered by — the ZoikoAvail™ answer rendered as a structured, auditable card rather than prose. **Availability is never asserted in free text alone.**

```
|   You: Is amoxicillin 500mg
available      |
|             near Nakuru?
|
|
|
|   Zoi:
|
|
|_____
|   |
|   | AMOXICILLIN 500MG CAPSULES
|   |
|   | Nakuru region
```

```

| |
| |
| |
| | • Available – High confidence
| |
| | 12 stocking pharmacies
| |
| |
| |
| | Data as of 14:32 EAT .
ZoikoAvail™ | |
| |
| |
| | [ View pharmacies ] [ Set
alert ] | |
|
|_____
|_____
|
| Confidence reflects pharmacy-
reported |
| and signal-derived data. Stock
can |
| change quickly – the alert will
notify |
| you of changes.
|

```

Card contract (non-negotiable):

- Confidence tier (High / Moderate / Low) mapped directly from the ZoikoAvail™ Bayesian model – never a raw percentage (false precision) and never omitted.
- Timestamp + data-source attribution on every card.
- Below the model's confidence floor, the card renders an honest "insufficient signal" state with the alert CTA – Zoi never guesses.
- Card states: available · limited · unavailable · insufficient-signal · stale-data (>N hours old, visually flagged).

3.5 Guardrail state: medical-advice refusal + human handoff

```
|   You: Can I take this with  
warfarin?      |  
|  
|  
|   Zoi:  
|
```

| That's an important question, and
it's |
| one for a clinical professional –
drug |
| interactions depend on your
specific |
| situation, and I'm not able to
advise |
| on them safely.
|
|
|
| Your pharmacist can answer this
when |
| you collect your medicine. If
you'd |
| like, I can show you the stocking
|
| pharmacies' contact details now.
|
|
| [Show pharmacy contacts]
|
| [Continue with availability]
|

Rules: refusal is warm, explains *why* in one clause, and always converts into a legitimate

next step. No "I'm just an AI" self-deprecation;
no lecture. Enforced by a dedicated safety
classifier ahead of the response model (§6),
not prompt-only.

3.6 Contextual entry variants (page-aware launcher)

Entry page	Launcher label	Skips to
Platform	◆ Ask Zoi	Persona routing
Patients	◆ Find a medicine	Availability flow, patient persona
Pharmacies	◆ Ask Zoi	Pharmacy onboarding flow
Enterprise	◆ Talk to us	Enterprise qualification → human routing
Wholesale Portal	◆ Partner support	Authenticated wholesale flow
Login	◆ Need help	Auth-support flow (no


```
|   □ Include this conversation
|   ← consent, default ON,
|
|   genuinely optional
|   [ Request contact ]
|
|
|
|   — after submit —
|
|   Request received – reference #ZK-
4821. |
|   The team responds within one
business |
|   day.
|
```

Escalation triggers: explicit user request · two consecutive low-confidence responses · any safety-classifier hit beyond availability scope · enterprise/wholesale commercial intent (routed to sales queue, not support).

3.8 Error and degraded states

```
Offline / API failure:
|   Zoi is temporarily unavailable.
|
```

```
|   For urgent needs:
support@zoikomeds...   |
|   [ Retry ]
|

ZoikoAvail™ degraded (chat up, data
plane down):
|   I can talk, but live availability
data   |
|   is temporarily unreachable. I can
set     |
|   an alert that fires once it's
back.   |
```

Rule: errors state what happened and the next step. No apologising avatars, no "Oops!".

4. Conversation Design System

Voice: precise, calm, humane. Sentence case. No exclamation marks. No emoji. Contractions allowed (warmth without informality). Second person always.

Message rules:

- Zoi's turns ≤ 3 short paragraphs; structured data always goes to cards, never prose.
- Maximum 3 suggestion chips per turn; chips are verbs ("Set alert," "View pharmacies"), never questions back to the user.
- One question per turn, maximum.
- Latency: streaming tokens begin $< 1.5s$ p95; a static "Zoi is thinking" indicator (no bouncing dots) appears at 400ms.

Localization: copy externalized from day one (ICU MessageFormat); confidence-tier labels and disclosure lines are legal-reviewed strings per locale, not free translations.

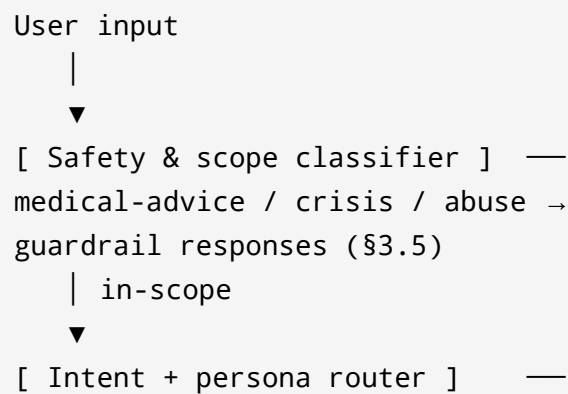
5. Accessibility (WCAG 2.2 AA — floor, not target)

- Panel is a `role="dialog"` with focus trap; launcher is a `<button>` with `aria-expanded`.
- Full keyboard operation: open (launcher), Esc to close, arrow-key navigation across

chips and card actions.

- Live region (`aria-live="polite"`) for incoming messages; streaming buffered into sentence-level announcements to avoid screen-reader flooding.
- Confidence tiers encoded by label + shape, never color alone.
- Respects `prefers-reduced-motion` (no slide/scale animation), 200% zoom without horizontal scroll, 4.5:1 minimum contrast throughout.

6. AI Architecture (Global Head of AI)



```
availability / navigation /  
onboarding / commercial / support  
|  
▼  
[ Grounding layer – RAG ]  
  └─ ZoikoAvail™ API      →  
    availability queries (structured,  
    never free-recall)  
    └─ MediBase™ resolver  →  
      medicine-name disambiguation  
      (brand/generic/strength)  
    └─ Site content index  →  
      platform, compliance, onboarding  
      answers  
    └─ Nothing else        → out-  
      of-corpus questions get honest "I  
      don't know" + escalation  
    |  
    ▼  
    [ Response model ] → streamed, card-  
    structured output  
    |  
    ▼  
    [ Output validator ] → schema check  
    on cards; blocks any availability  
                                assertion  
    lacking source + confidence +  
    timestamp
```

Non-negotiable AI principles:

- 1. The model never recalls availability from parameters.** Availability exists only as a tool result; the output validator rejects availability language without an attached ZoikoAvail™ payload.
- 2. Confidence is inherited, not generated.** The card's tier is a deterministic mapping from the ZoikoAvail™ Bayesian posterior — the LLM cannot inflate it.
- 3. Defense in depth.** Safety classifier (pre) + system policy (in) + output validator (post). Prompt-only guardrails are not acceptable in a healthcare domain.
- 4. Full audit trail.** Every turn logged with classifier verdicts, retrieved sources, tool payloads, and validator outcomes — retained per the privacy engineering stack, PII-minimized, and referenced from the Trust Center.
- 5. Evaluation before launch.** A red-team suite covering medical-advice elicitation, availability hallucination, persona confusion, and prompt injection;

regression-gated on every model or prompt change.

7. Engineering Specification (Global CTO)

Component hierarchy:

```
<ZoiProvider>           // session,
persona, feature flags, consent state
├─ <ZoiLauncher/>       // page-
aware label logic
└─ <ZoiPanel/>           // dialog,
focus management
    ├─ <ZoiHeader/>
    ├─ <ZoiViewport/>    //
virtualized message list
    │   ├─ <MessageText/>
    │   └─ <AvailabilityCard/> //
schema-validated render
    │   ├─ <EscalationForm/>
    │   └─ <GuardrailNotice/>
    ├─ <ChipRow/>
    └─ <ZoiComposer/>    // input,
send, rate limiting
```

Contracts and NFRs:

- Transport: SSE streaming over the existing API gateway; WebSocket unnecessary at this interaction pattern.
- AvailabilityCard payload is a versioned JSON schema (`zoi.card.v1`) shared between output validator and client – the client renders only validated payloads and drops malformed ones to a text fallback.
- Session: anonymous by default (ephemeral session ID); authenticated context only inside Wholesale/logged-in surfaces, respecting the NIST 800-63B AAL boundaries already specified for Login. **Zoi never displays account or order data to an unauthenticated session.**
- Residency: conversation data stays within the user's regional cell (cell-based architecture); no cross-region log aggregation of raw transcripts.
- Performance budget: launcher JS $\leq$ 8KB gzipped, lazy-load panel bundle on first open; zero impact on core-page LCP.

- Availability: chat degrades independently of the data plane (§3.8); circuit breaker on ZoikoAvail™ calls with the honest degraded state, never cached-as-fresh answers.
 - Telemetry: open rate, persona distribution, containment rate, escalation rate, guardrail-hit rate, card CTR, p95 first-token latency.
Explicitly not tracked: message content analytics on patient conversations beyond aggregate intent classes.
-

8. Rollout

1. **Phase 1 — Grounded navigator.** Site-content RAG + persona routing + escalation. No availability tool. Establishes voice, guardrails, telemetry baseline.
2. **Phase 2 — Availability.** ZoikoAvail™ tool + Availability Card + alerts. The red-team suite gates this phase.
3. **Phase 3 — Authenticated surfaces.** Wholesale Portal support flows, order-status queries behind AAL2.

4. Phase 4 – Proactive intelligence.

ZoikoSignal™-driven alert conversations (shortage early-warning delivered through Zoi, opt-in).

9. Governance Dependencies (blocking)

1. **Trust Center (unbuilt – previously flagged as the platform's most critical gap).** The persistent footer ("Zoi is AI... Learn how →"), the AI disclosure policy, the conversation-data retention statement, and the audit posture all resolve to Trust Center anchors. **Zoi cannot ship Phase 1 to production before the Trust Center exists.**

This is now the second major asset blocked on it – the case for building the Trust Center next has strengthened, not weakened.

2. **Legal review** of disclosure strings, refusal copy, and per-locale confidence-tier language.

3. **Naming clearance** for Zoi™ within the Zoiko Group trademark family across target jurisdictions.

End of specification. Next recommended session: the Trust Center page – now blocking two shipped-quality assets.